

2D Wind Tunnel in Rust

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For my project I'll be implementing the Lattice Boltzmann Method to build a 2D Wind Tunnel that can be used to measure air resistance, lift, etc.

The program can be broken into the following pieces:

1. Physics calculations
2. Interface for building an object to test
3. Visual representation of the fluid simulation
4. Controls for affecting parameters of the fluid simulation

Physics Calculations

- Using LBM D2Q9 with BGK collision

The collision step

For the Bhatnagar Gross and Krook (BGK)^[3] model, which leads relaxation to equilibrium via collisions between the molecules of a fluid, we have

$$f_i^*(\vec{x}, t) = f_i(\vec{x}, t) + \frac{f_i^{eq}(\vec{x}, t) - f_i(\vec{x}, t)}{\tau_f},$$

where $f_i^*(\vec{x}, t)$ is the new lattice density, and $f_i^{eq}(\vec{x}, t)$ is the equilibrium density along direction i which can be expressed by using a Taylor expansion (see below, in [Mathematical equations for simulations](#)):

$$f_i^{eq} = \omega_i \rho \left(1 + \frac{3\vec{e}_i \vec{u}}{c^2} + \frac{9(\vec{e}_i \vec{u})^2}{2c^4} - \frac{3(\vec{u})^2}{2c^2} \right).$$

The model assumes that the fluid locally relaxes to equilibrium over a characteristic timescale τ_f . This timescale determines the [kinematic viscosity](#), the larger it is, the larger is the kinematic viscosity.

The streaming step

$$f_i(\vec{x} + \vec{e}_i \delta_t, t + \delta_t) = f_i^*(\vec{x}, t)$$

As $f_i^*(\vec{x}, t)$ is, by definition, the fluid density at point $\vec{x}$ at time t , that is moving at a velocity of $\vec{e}_i$ per time step, then at the next time step $t + \delta_t$ it will have flowed to point $\vec{x} + \vec{e}_i \delta_t$.

- have to figure out how to make this into actual rust code

Design Interface

- Edit very small png files pixel by pixel
- allow usage of your own templates
- will need to decide sizing based on performance but will start small

Visual Representation

- Will use the Crate Pixels
- <https://docs.rs/pixels/latest/pixels/>

Parameter Controls

- Ability to change:
 - Simulation speed
 - Inlet speed
 - A few presets for “fluid type” which have values for viscosity and density

inspiration:

https://www.reddit.com/r/CFD/comments/1gfnkg3/rust_lbm_solver_on_the_gpu/
<https://dokumen.live/reviews/s5l5Al/246485/4988549-fluid-dynamics-for-dummies>

References (so far):

<https://www.fidelisfea.com/post/navier-stokes-vs-lattice-boltzmann-for-cfd-a-comparative-analysis>

<https://www.oatext.com/lattice-boltzmann-modeling-for-mass-and-velocity-fields-of-casting-flows.php>

<https://users.rust-lang.org/t/announcing-pixels-hardware-accelerated-pixel-frame-buffer/34326>